FireFusion Capstone Project

Aspect Dataset Pipeline
Technical Documentation

Prepared for tutor review and team handoff

Report purpose
This document briefly explains the dataset choice, workflow, outputs, validation, and handoff files for the aspect contribution in FireFusion. The write-up follows the FireFusion data engineering guideline and is kept short for quick review.

Document Contributor	Abdul Suboor Mohammed
Project stream	FireFusion - Data Engineering
Individual contribution	Aspect dataset pipeline

1. Project context and scope

FireFusion is building a data engineering workflow for bushfire-related data that can support downstream analytics, AI modelling, and dashboard work. My individual task was to contribute one topography-derived feature without overlapping with teammates already working on elevation and slope, so I focused on aspect.

The current work is a small working pipeline on a sample extent. It is meant to show the full flow clearly before any larger-scale expansion.

2. Objective of my task

Question	Answer
What did I build?	A small aspect pipeline that downloads DEM data, generates an aspect raster, exports a CSV sample, and validates the output.
Why aspect?	It complements the other topography features and avoids duplicate work in the team.
Why is this useful?	Aspect shows terrain direction and can later be joined with fire, weather, and risk data.
What shows the task is complete?	Working scripts, sample outputs, validation report, schema draft, README, and documented handoff files.

3. Dataset and tools used

The workflow uses SRTM1 DEM data downloaded programmatically with the elevation/eio utility. The aspect raster is then generated from the DEM using GDAL's gdaldem aspect tool.

Component	Used in this task	Purpose
SRTM1 DEM	Yes	Base elevation raster used as the input surface for aspect generation.
eio / elevation	Yes	Downloads and clips SRTM data for a chosen bounding box.
GDAL - gdaldem aspect	Yes	Creates the derived aspect raster from the DEM.
Rasterio + NumPy	Yes	Reads raster output, handles nodata, and prepares sample extraction.
CSV export	Yes	Creates a structured sample output for downstream ETL or handoff.
JSON validation report	Yes	Stores raster and CSV checks in a simple handoff format.
PostgreSQL schema draft	Yes	Shows how aspect data could be stored later by the backend/database stream.

Source information: NASA SRTM data via the elevation/eio utility. Collection method: programmatic download using a sample bounding box.

4. Workflow

- Step 1 - Define sample extent: a small bounding box is chosen to test the pipeline before scaling it further.
- Step 2 - Extract DEM input: the data script runs eio and saves the clipped DEM as dem_sample.tif.
- Step 3 - Transform to aspect: the pipeline runs gdaldem aspect and saves aspect_sample.tif.
- Step 4 - Clean raster values: nodata values are converted to NaN so they do not affect the checks.
- Step 5 - Load sample output: sample longitude, latitude, and aspect_deg rows are exported to aspect_sample_points.csv.
- Step 6 - Validate results: CRS, bounds, raster shape, valid range, CSV row count, and duplicate rows are checked.
- Step 7 - Prepare handoff files: the JSON report, SQL schema draft, and README are created for reuse by the team.

Pipeline summary: Extract -> Transform -> Validate -> Load (CSV/sample output) -> Handoff files

5. Scripts, outputs, and folder structure

5.1 Script responsibilities

File	Main role	Key result
data_file.py	Downloads the sample DEM using eio and stores it in the data folder.	dem_sample.tif
pipeline_file.py	Generates the aspect raster, extracts sample points, performs validation, and visualises the raster.	aspect_sample.tif, aspect_sample_points.csv
aspect_validation_report.json	Stores raster and CSV checks in a machine-readable format.	Validation summary for handoff

aspect_schema.sql	Defines a draft database table for future integration.	Backend-facing storage draft
README.md	Explains purpose, workflow, and outputs.	Quick team handoff reference

5.2 Output files produced

- data/dem_sample.tif - input DEM used by the pipeline
- data/aspect_sample.tif - derived aspect raster
- data/aspect_sample_points.csv - sample structured output containing longitude, latitude, and aspect_deg
- reports/aspect_validation_report.json - validation report for raster and CSV checks
- schema/aspect_schema.sql - schema draft for possible PostgreSQL integration
- README.md - short handoff summary

5.3 Folder structure

```
aspect_pipeline/
├── notebooks/
│   └── aspect_pipeline.ipynb
├── scripts/
│   ├── data_file.py
│   └── pipeline_file.py
├── data/
│   ├── dem_sample.tif
│   ├── aspect_sample.tif
│   └── aspect_sample_points.csv
├── reports/
│   └── aspect_validation_report.json
├── schema/
│   └── aspect_schema.sql
└── README.md
```

This structure keeps scripts, outputs, and documentation separate so the pipeline is easier to review and reuse.

6. Validation and result interpretation

The output should be explained as a validated sample pipeline, not as a final production dataset. The aspect values are within an expected range, and the outputs are structured for further ETL use.

- The aspect raster was created successfully from the DEM input.
- The CSV export was produced successfully from raster samples.
- CRS, bounds, and raster dimensions were checked.
- NoData was handled before computing summary statistics.
- Aspect values were confirmed to fall within the expected 0 to 360 range after cleaning.
- The CSV output was checked for duplicate rows.
- A JSON validation report and SQL schema draft were created as handoff files.

All validation steps are reproducible and can be rerun from the raw DEM input using the provided scripts.

6.1 Explaining aspect in the interview

Aspect represents the direction a terrain surface faces. In simple terms, it shows slope orientation. Common examples are 0 degrees for north-facing, 90 degrees for east-facing, 180 degrees for south-

facing, and 270 degrees for west-facing terrain. The values in the raster therefore show terrain direction across the sampled area.

Note: the current implementation proves the pipeline works on a sample area. The same logic can later be extended to larger extents or replaced with the final team-agreed topography source if needed.

7. Data contract and handoff

- Output: aspect_sample_points.csv
- Fields: longitude, latitude, aspect_deg
- Grain: point-level terrain orientation
- Usage: can later be joined with fire, weather, or risk datasets
- Known limitation: the current implementation uses a sample extent only

8. Challenges faced and how they were solved

Issue	What happened	Resolution
Large-scope confusion	The initial discussion included wide-area topography ideas that were too heavy for a quick MVP.	The task was narrowed to an individual aspect pipeline built on a sample extent.
Data-source access friction	Some portals provided metadata packages or delayed access instead of an immediately usable raster.	The implementation switched to a directly accessible SRTM-based Python workflow.
GDAL dependency issues	The elevation package required GDAL command-line utilities to be installed correctly.	GDAL tools were installed and the working CLI-based approach was used.
Environment compatibility	There was a NumPy/pandas compatibility error while exporting CSV.	The workflow was simplified during the export step and then stabilised.
Path-with-spaces error	The download command failed when the working directory path contained spaces.	The project was moved to a clean path without spaces so the CLI tools could run correctly.

9. Short speaking script

For my FireFusion contribution, I focused on the aspect dataset as part of the topography stream. I used a sample SRTM DEM as the input surface, built a small pipeline that downloads the DEM, derives the aspect raster using GDAL, and exports sample values into CSV. I then validated the raster using CRS, bounds, shape, NoData handling, and aspect-range checks, and I prepared a JSON report, SQL schema draft, and README so the work is easier for team handoff. This gives the project a working MVP-style aspect pipeline that can later be scaled or integrated with the broader ETL workflow.

Appendix A. Evidences

Figure A1. Data folder output evidence

Screenshot showing dem_sample.tif, aspect_sample.tif, and aspect_sample_points.csv.

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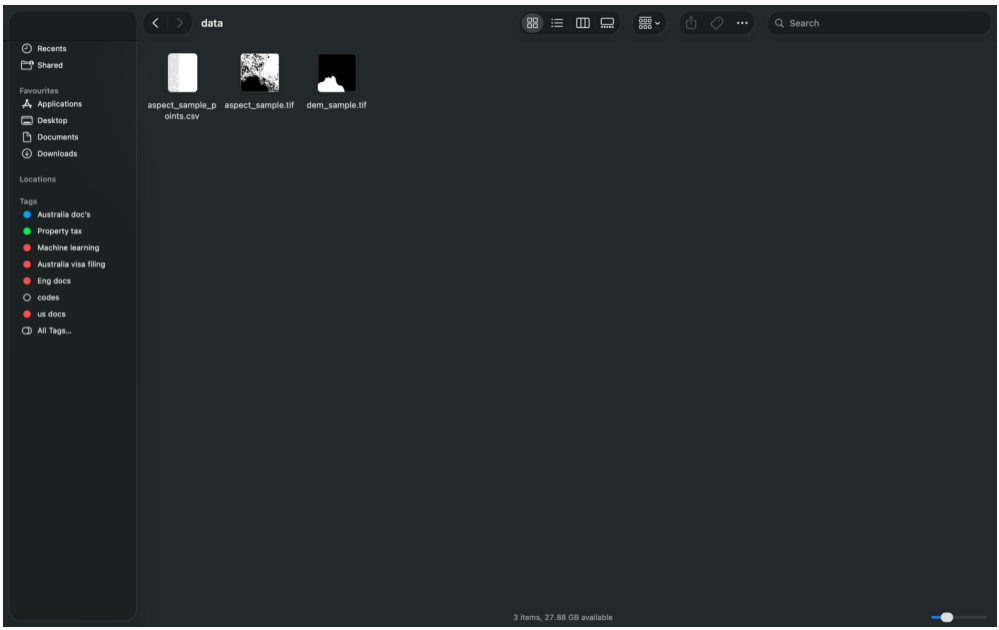


Figure A2. DEM download evidence

Notebook or terminal screenshot confirming DEM download success.

Figure A3. Aspect generation evidence

Paste the notebook or terminal screenshot showing the aspect raster creation command/output.

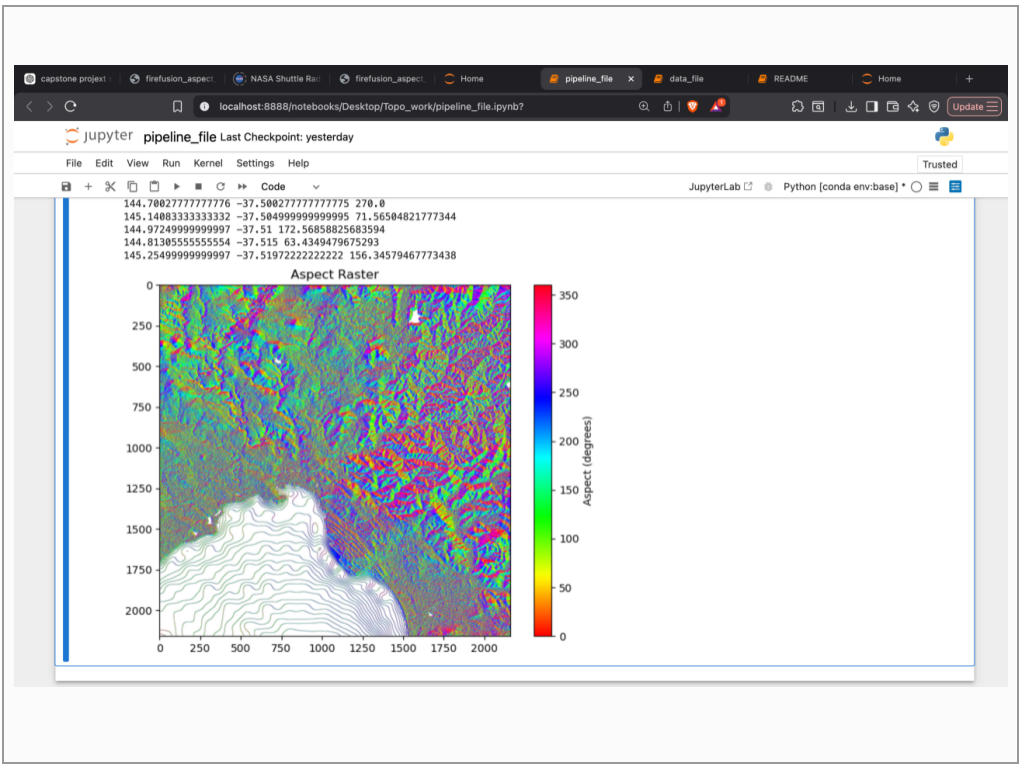


Figure A4. Raster validation evidence

Paste the screenshot that shows CRS, bounds, raster shape, min, and max aspect values.

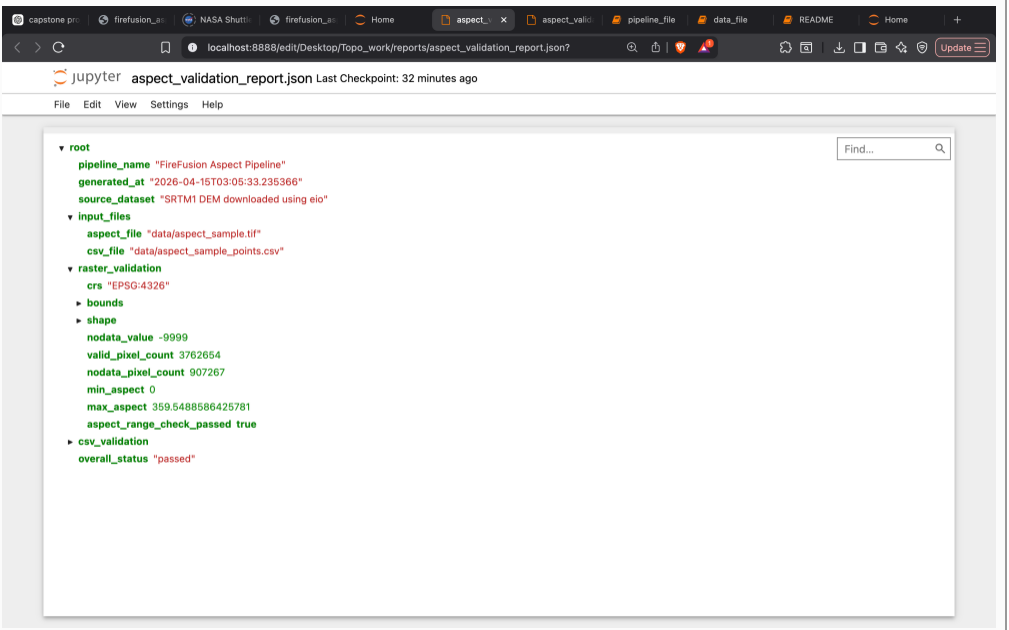
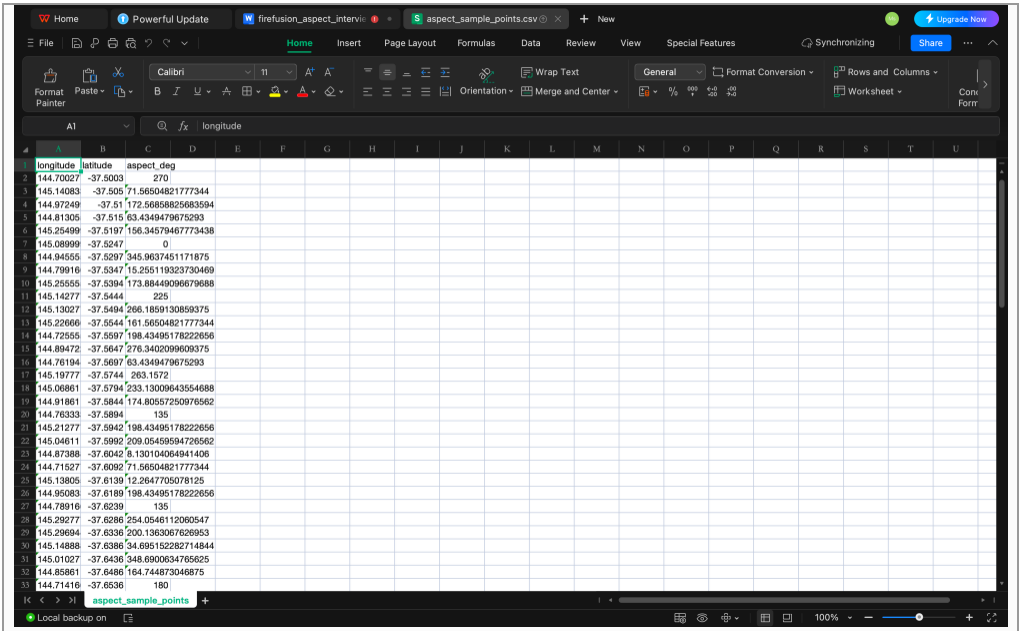


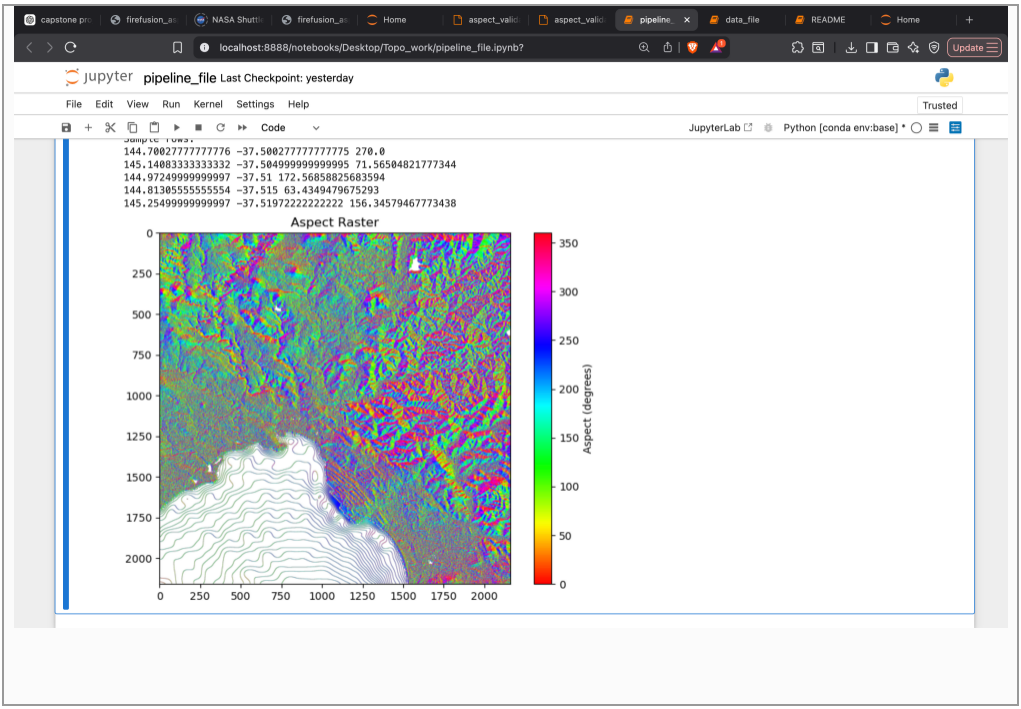
Figure A5. CSV evidence Tthe screenshot that shows sample rows from aspect_sample_points.csv or the CSV file opened in a viewer.



longitude	latitude	aspect_deg
144.70027	-37.5003	270
145.14083	-37.505	71.56504821777344
144.97249	-37.51	172.56858825683594
144.81305	-37.515	63.4349479675293
145.25499	-37.5197	156.34579467773438
145.08999	-37.5247	0
144.94555	-37.5297	345.9637451171875
144.79916	-37.5347	15.255119323730469
145.25555	-37.5394	173.88449096679688
145.14277	-37.5444	225
145.13027	-37.5494	266.1859130859375
145.22666	-37.5544	161.56504821777344
144.72555	-37.5597	198.43495178222656
144.89472	-37.5647	276.3402099609375
144.76194	-37.5697	63.4349479675293
145.19777	-37.5744	263.1572
145.06861	-37.5794	233.13009643554688
144.91861	-37.5844	174.80557250976562
144.76333	-37.5894	135
145.21277	-37.5942	198.43495178222656
145.04611	-37.5992	209.05459594726562
144.87388	-37.6042	8.130104064941406
144.71527	-37.6092	71.56504821777344
145.13805	-37.6139	12.2647705078125
144.95083	-37.6189	198.43495178222656
144.78916	-37.6239	135
145.29277	-37.6286	254.0546112060547
145.29694	-37.6336	200.1363067626953
145.14888	-37.6386	34.69515228271484
145.01027	-37.6436	348.6900634765625
144.85861	-37.6486	164.74487304687 5
144.71416	-37.6536	180

Figure A6. Final aspect map

Final visual plot of the aspect raster from notebook.


localhost:8888/notebooks/Desktop/Topo_work/pipeline_file.ipynb?
jupyter pipeline_file Last Checkpoint: yesterday
File Edit View Run Kernel Settings Help
Trusted
Code
JupyterLab
Python [conda env:base]
144.70027777777776 -37.50027777777775 270.0
145.14083333333332 -37.504999999999995 71.56504821777344
144.97249999999997 -37.51 172.56858825683594
144.81305555555554 -37.515 63.4349479675293
145.25499999999997 -37.51972222222222 156.34579467773438
Aspect Raster
Aspect (degrees)